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ABSTRACT

abstract

Keywords: word1 — word2 — word3 — word4 — word5

1. INTRODUCTION

The PageRank algorithm, first introduced by Larry Page and Sergey Brin, was one of the first algorithms to rank webpages based on their link structure rather than just content Brin & Page (1998). The algorithm models the web as a directed graph and simulates a "random surfer" who follows hyperlinks with some probability or jumps to a random page to avoid getting stuck Page et al. (1999).

This setup defines a Markov process where each page's importance is proportaional to the importance of pages linking to it. The final PageRank vector corresponds to the steady-state distribution of this process and can be found either by iteration or by solving for the dominant eigenvector of the hyperlink matrix $\boldsymbol{H}$ Langville & Meyer (2009).

In this project, we first implemented PageRank using the difference equation $x_{k+1} = \boldsymbol{H}x_k$, which resulted in the scores converging once the solution reached the maximum tolerance allowed. We ten confirmed our result by solving for the eigenvector associated with the largest eigenvalue of the matrix $\boldsymbol{H}$. Our goal was to understand how the algorithm behaves under different network structures and how quickly it converges Langville & Meyer (2004).

2. DATA AND OBSERVATIONS

To start, we defined a 6-node directed graph where each page links to one or more pages, as shown in Figure 1. From this, we constructed the hyperlink matrix $\boldsymbol{H}$, where each non-zero entry $\boldsymbol{H}_{ij}$ is equal to $\frac{1}{|P_{ij}}$, the reciprocal of the number of outbound links from page P_j. This matrix represents the probability that a random surfer on page j clicks through to page i:

$$\boldsymbol{H} = \begin{pmatrix} 0 & 0 & 1/3 & 0 & 0 & 0 \\ 1/2 & 0 & 1/3 & 0 & 0 & 0 \\ 1/2 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/2 & 1 \\ 0 & 0 & 1/3 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 1/2 & 1/2 & 0 \end{pmatrix} \tag{1}$$

We initialized the PageRank vector $\boldsymbol{x}_0$ with equal values and assumed that each page is equally likely to be visited from the start:

$$\boldsymbol{x}_0 = \begin{pmatrix} 1/6 \\ 1/6 \\ 1/6 \\ 1/6 \\ 1/6 \\ 1/6 \end{pmatrix} \tag{2}$$

We then applied the iterative update rule $\boldsymbol{x}_{n+1} = \boldsymbol{H}\boldsymbol{x}_n$ until the rank vector converged below a given tolerance to simulate a random surfer moving through the web graph following links. We visualized the convergence of the PageRank vector by plotting each page's rank score at each ieration, which allowed us to track how the scores changed until they reached a steady state. Our results are shown in Figure 2.

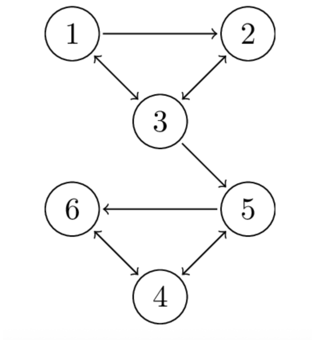

Figure 1. example graph 1.

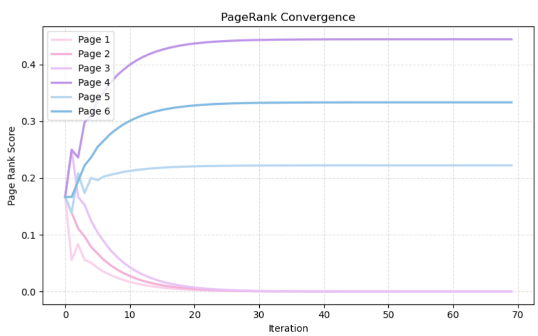

Figure 2. PageRank convergence for graph 1.

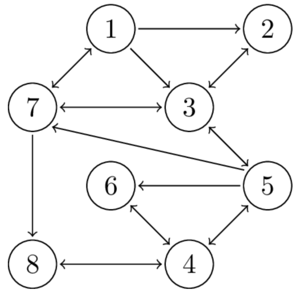

Figure 3. example graph 2.

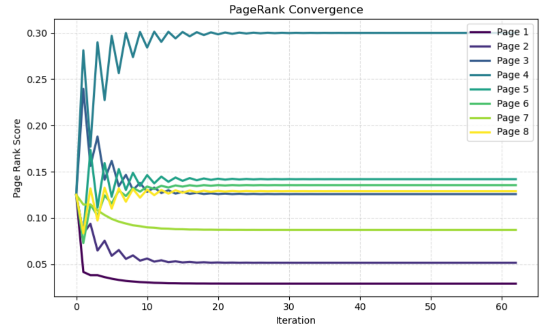

Figure 4. PageRank Convergence for graph 2.

3. RESULTS

results

4. SUMMARY AND CONCLUSION

summary/concl

4.1. *Acknowledgements*

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